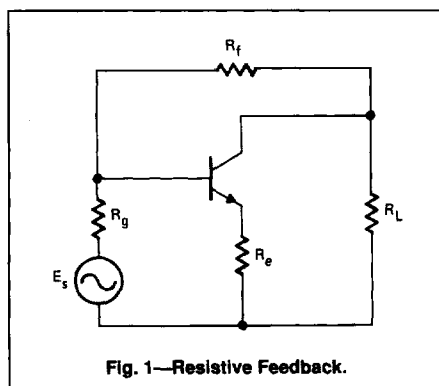


Application Information for Thin Film Cascadable Amplifiers



INSIDE THE TO-8 AMPLIFIER: BASIC CIRCUIT CONCEPTS

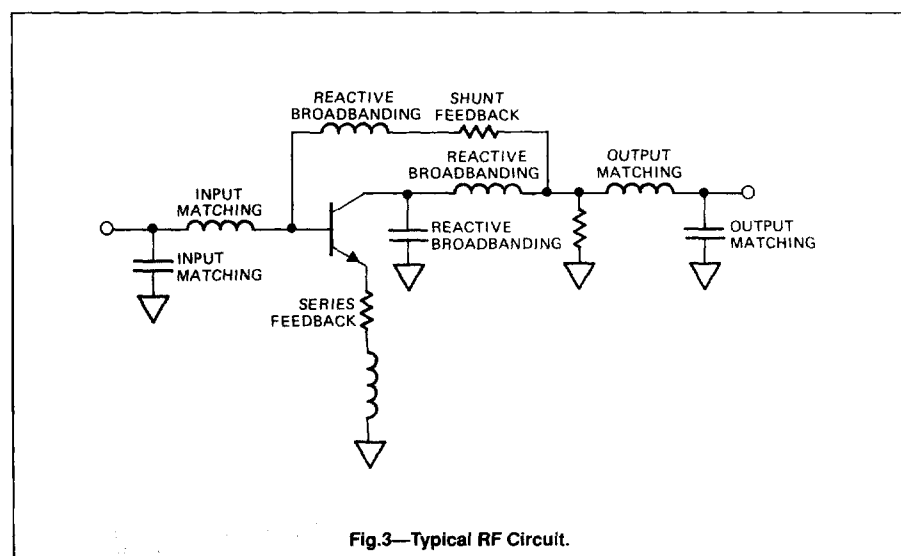
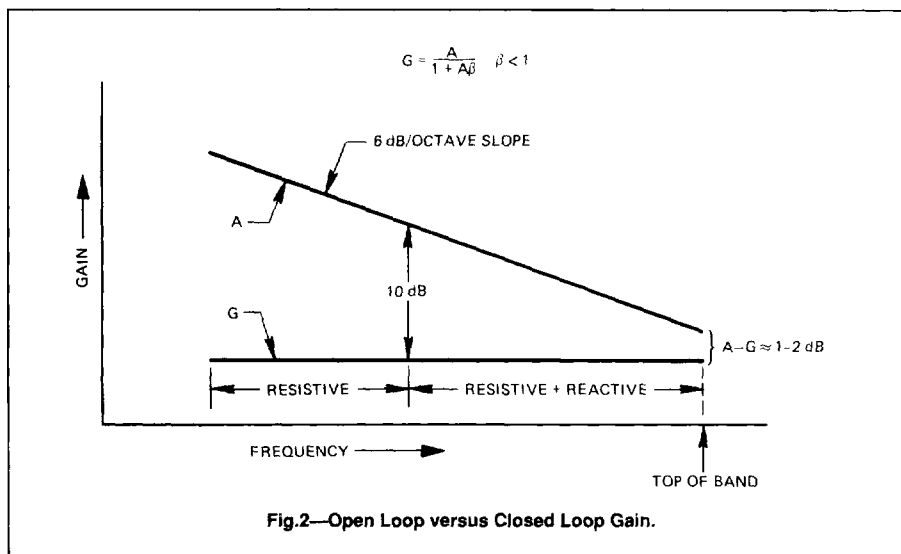
This discussion covers the basic circuit concept behind the cascable amplifier and details some of the techniques that are used to maintain hardware stability in the application of the units. The fundamental common denominator to most broadband cascable amplifiers is collector-to-base, shunt-resistive feedback and series-emitter resistive feedback, as shown in Figure 1. There are some reactive feedback methods that have been employed both in combination with and without resistive feedback to reduce the degradation that the resistive feedback has on the noise figure. However, the consideration of these design schemes is beyond the scope of this article.



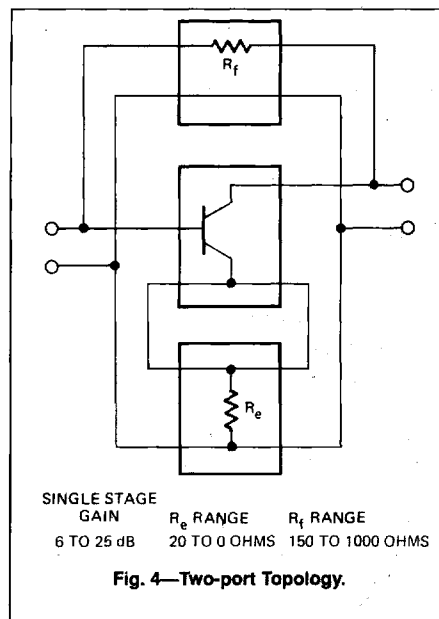
The typical transducers used in the TO-8s are microwave bipolar transistors which have a nominal 6 dB/octave gain slope, as shown in Figure 2. The amplifier gain curve can be flattened using pure, resistive feedback with no reactive elements up to a frequency where the open loop gain is approximately 10 dB greater than the closed

loop gain. Then, reactive elements must be used to provide input and output match and to "gain-peak" the amplifier to obtain the highest-gain bandwidth product. A

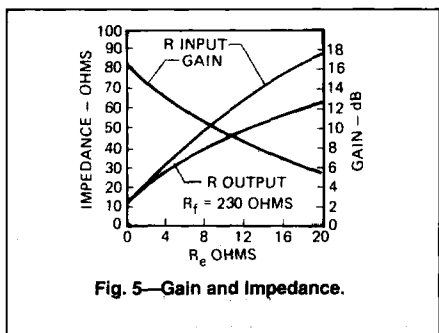
typical RF circuit excluding DC biasing, coupling, and bypass capacitors is shown in Figure 3. The parasitic elements, such as header capacitance and bondwire induc-



This program uses conventional matrix techniques to treat each circuit element and uses fixed topology. Single stage gain typically runs from 6 to 25 dB, with R_e ranging from 20 to 0 ohms and R_f ranging from 150 to 1000 ohms, as illustrated in Figure 4. For low noise circuits, R_e usually goes to zero, since the transistor is normally biased for less than 10 mA collector current and the internal R_e , which can be approximately $.026/I_c$, acts as sufficient emitter feedback. The curves in Figure 5 show the gain, the input and output impedance as a function of R_e with a fixed R_f . R_e is fixed and R_f varied in Figure 6, which



shows the inverse gain relationship between these two resistors. It is this property of the microwave bipolar transistor that allows various gain options to be realized. For a fixed value of R_E , to obtain as close to a 50-ohm input and output impedance as possible, a certain value of R_f



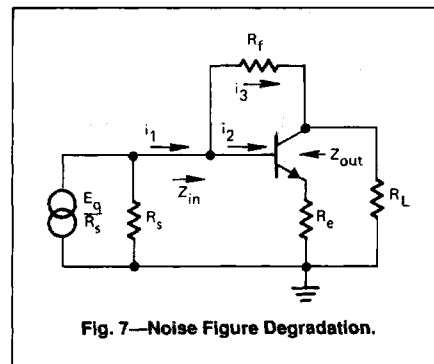
The graph shows the relationship between collector-base resistance (R_f) and both input impedance (R_{in}) and gain (A_v). The x-axis represents R_f in ohms, ranging from 100 to 600. The left y-axis represents impedance in ohms, ranging from 20 to 120. The right y-axis represents gain in dB, ranging from 2 to 14. Two curves are plotted: R_{in} (input impedance) and A_v (gain). Both curves show an increasing trend as R_f increases. The R_{in} curve starts at approximately 45 ohms at $R_f = 100$ ohms and reaches about 115 ohms at $R_f = 600$ ohms. The A_v curve starts at approximately 2.5 dB at $R_f = 100$ ohms and reaches about 13.5 dB at $R_f = 600$ ohms.

R_f (OHMS)	R_{in} (OHMS)	A_v (dB)
100	45	2.5
200	65	4.5
300	85	6.5
400	100	8.5
500	110	10.5
600	115	13.5

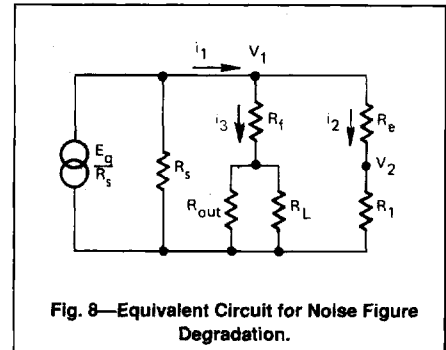
Noise-Figure Degradation

For a rough first approximation of the quantitative degradation in noise figure that the resistive feedback elements cause, the following derivations may be used. The approximation excludes the actual transistor and its noise properties from the analysis. This contributes to the degree of accuracy of the approximation; but, its purpose is to obtain, by simple calculation, what degradation can be expected. This type of approximation is particularly useful when it is desired to determine the influence of changing the feedback elements on noise figure to obtain different gain levels.

When the gain of the bipolar transistor is significantly reduced by feedback, its open-loop noise characteristics change such that the amplifier does not respond as readily to changes in source impedance. The transistor will generally operate better



The equivalent circuit for the transistor circuit in Figure 7 can be changed to that shown in Figure 8 for determining the



resistive losses caused by the feedback elements. In effect, the emitter feedback resistor, R_e , can be considered a loss element in series with R_1 , the effective input resistance of the transistor. The emitter resistor can be treated as a voltage-dividing element where the signal-to-noise ratio is increased by this voltage division.

Letting $Z_{in} = R_{out} = R_L = 50$ ohms
and solving for R_1 ,

$$R_1 = \frac{(R_f + 25) 50}{R_f - 25} - R_e \quad (1)$$

The input power to the network when $Z_{in} = 50 \text{ ohms}$ is $50i_1^2$. The power delivered to R_1 is $R_1(i_2)^2$. Taking the ratio of delivered power to input power yields:

$$\text{Power ratio} = \frac{R_1 (i_2)^2}{50(i_1)^2} \quad (2)$$

Using current division, and solving for i_2 , gives:

$$i_2 = \frac{i_1 (R_f + 25)}{R_f + 25 + R_e + R_1} \quad (3)$$

Substituting equation 3 into equation 2 yields:

$$\text{Power ratio} = 10 \log \frac{(R_f + 25)^2 R_1}{50(R_f + 25 + R_e + R_1)^2} \quad (4)$$

For an amplifier with an R_e of 12 ohms and R_f of 240 ohms, the degradation of the noise figure calculates to 1.85 dB. At the other end of the scale a low noise amplifier might have an R_f of 400 ohms and an R_e of zero ohms, from which only 0.54-dB degradation is calculated.

Empirical data show that using equations 1 and 4 can be made slightly more accurate by adding a correction factor of 0 dB for a 20-dB or higher gain stage and 0.5 dB for a 7.5-dB gain stage, with linear interpolation in between.

Output Power

The output power of the TO-8 feedback amplifier is a direct function of the collector current. The great majority of cascable amplifiers use direct capacitively coupled outputs, i.e., circuits without impedance transforming devices and, as a result, are voltage-limited. Nonsymmetrical clipping occurs on the cutoff portion of the AC swing, as shown in Figure 9. The steep AC load line is a result of the direct coupling to the 50-ohm load impedance. For greater output power at a given collector current, the equivalent AC load impedance must be raised. This reduces the slope of the AC load line, thereby providing greater AC swing in the cutoff part of the cycle. This effect can be observed by comparing V_p to V_p' in Figure 9. For wideband operation and good impedance matching, it is generally not practical to use a transforming device, except for the VHF and lower UHF frequency ranges. The WJ-PA2 A87 series, A81 / A82 series and the

A67 series use autoformers in the collector circuit for higher and more efficient output power, but are limited to below-800-MHz operation. Because of the sensitivity of the output power to the AC load line, the voltage limiting is a direct function of the load impedance. Higher than 50 ohms for the load will result in greater than the nominal output power, and lower than 50 ohms will reduce the available output power.

DC Biasing

Depending on RF requirements and cost considerations, several schemes are used to provide the DC biasing in the cascable amplifiers. In the low-cost scheme shown in Figure 10, the collector feedback resistor serves the dual purpose of providing base bias and collector-to-bases hunt feedback. This approach is used in TO-5 packages and is represented by the W-J EA series, such as the EA2 amplifier. Compromised performance results because the ultra-small size does not allow input or output matching. The unit operates essentially with resistive matching, except that a series peaking inductor is used in the collector line. In addition, the value of R_1 is typically less than 150 ohms, which causes substantial noise figure

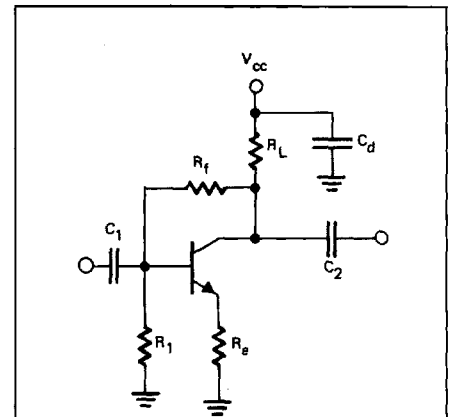


Fig. 10—Low-cost Bias: Compromised Performance.

degradation because the value of R_f must be chosen to set the gain and provide a reasonable input and output match. This requires that R_1 have a low enough value to enable it to pass the excess bleeder current. Since the compromised performance of the circuit shown in Figure 10 has no R_3 , in comparison to the circuit shown Figure 11, the voltage at the base will be low, which also necessitates that R_1 be low. In order to reduce the current flow, V_{CE} is set low, thereby reducing the maximum available gain in this type of design.

The passive bias circuit shown in Figure 11 overcomes noise figure and gain problems by using a separate R_f resistor and an emitter resistor to provide both DC feedback for temperature stability and to raise the voltage level at the base so that R_1 can be larger in value. The parallel combination of R_1 and R_2 provides a minimum amount of shunting and its accompanying noise figure increase and gain reduction.

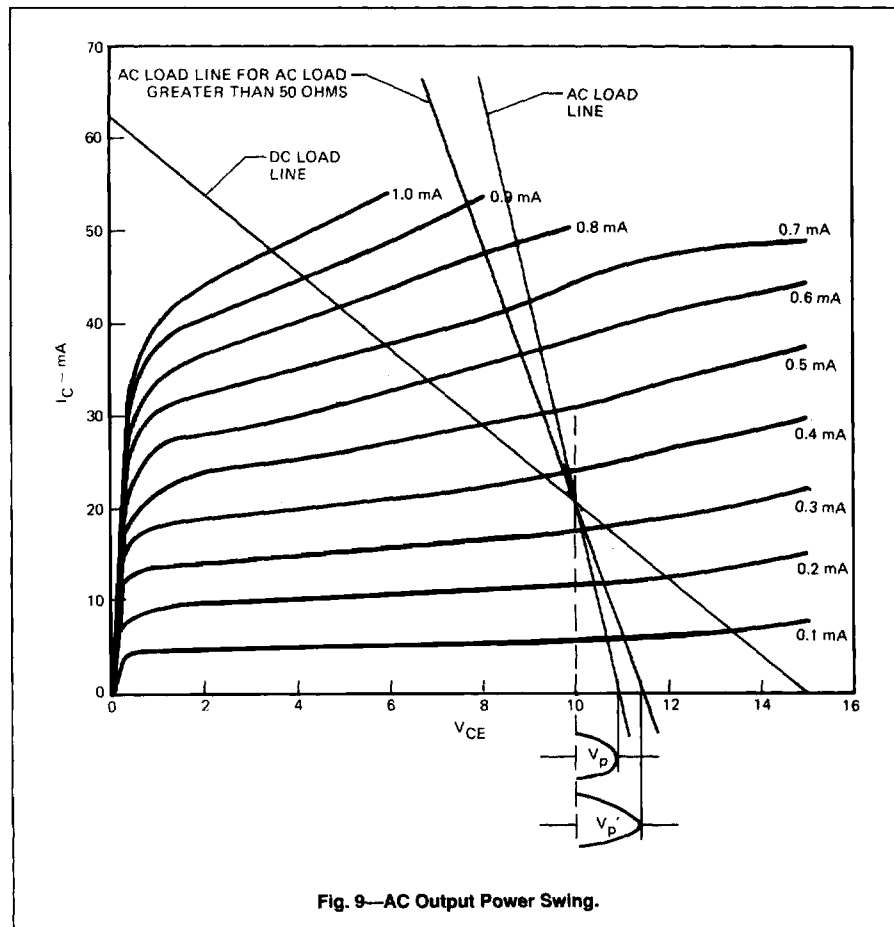


Fig. 9—AC Output Power Swing.

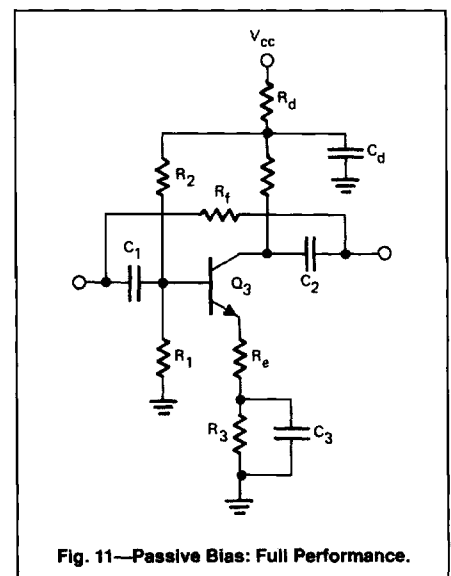


Fig. 11—Passive Bias: Full Performance.

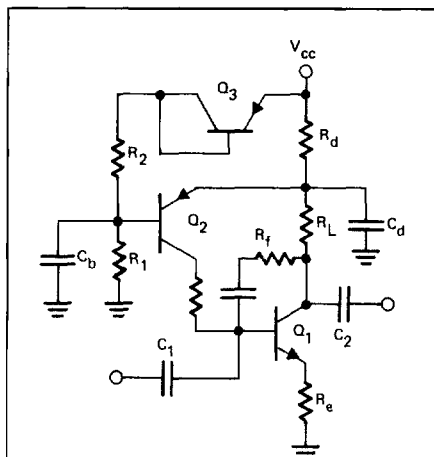


Fig. 12—Active Bias: Full Performance.

A third scheme uses a PNP low frequency transistor to bias the RF transistor, as shown in Figure 12. This approach virtually eliminates base shunting and, in addition, does not have an element in the emitter circuit which minimizes degeneration because of additional parasitic inductance in the emitter leg. This circuit is temperature-stable because if I_C in the RF transistor goes up, the voltage across the sense resistor, R_d , will go up, causing V_{BE} across Q_2 to go down. This action reduces the base drive to Q_1 , causing I_C to go down. When R_d is less than about 300 ohms, it is usually necessary to include Q_3 to track the base-emitter voltage of Q_2 over temperature as the differential voltages on R_d can become comparable in magnitude to the normal base-emitter voltage change that occurs as a function of temperature.

The circuit of Figure 13 is used for the higher-level cascable amplifiers, which differ from the type shown in Figure 12 by the addition of an RF choke. The choke

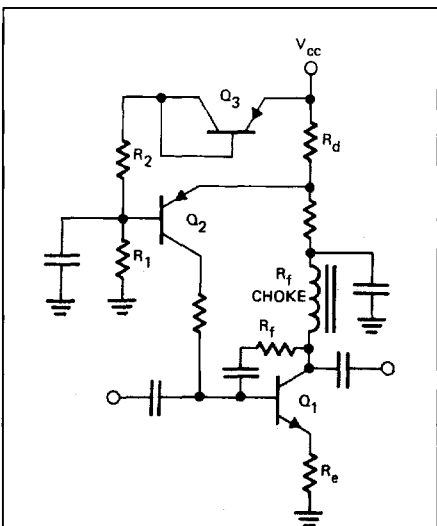


Fig. 13—Linear Power Amplifier: Active Bias.

does not limit current, and for higher-level units, particularly if it is not desired to raise V_{CC} above 15 volts, it does not shunt the output and absorb RF power. Most W-J amplifiers that provide +13 dBm or greater output power utilize an RF choke for power supply decoupling.

One method of providing bias for two cascaded RF stages in a single TO-8 is demonstrated in Figure 14. Here, the active bias circuit of Figure 12 is expanded to provide bias from a single transistor for both RF transistors. The limitation of this circuit is that the collector current of each RF transistor (Q_1 and Q_2) must be close enough in value so that each transistor will receive its proportionate amount of correcting currents from Q_3 when the circuit is operated over wide temperature ranges.

To summarize these bias schemes, the circuit in Figure 10 is used where minimum cost and size are the dominant factors. The circuit in Figure 11 is used where higher performance is desired, but the frequency of operation is usually limited to 500 MHz or below. The circuit will be less sensitive to sub-band oscillations (see section on sub-band oscillations), since no active device is used in the bias circuit.

The bias circuit shown in Figure 12 is not frequency-limited, and offers the maximum gain obtainable from the RF transistor. The circuit of Figure 13 is used where it is desired to obtain higher power than possible from the circuit of Figure 12. The circuit of Figure 14 minimizes the area required for high-performance applications when two stages in a single TO-8 package are used.

Performance Parameters Cascading Rules

The wide selection of amplifiers combined with the WJ-G1 gain control module, offers

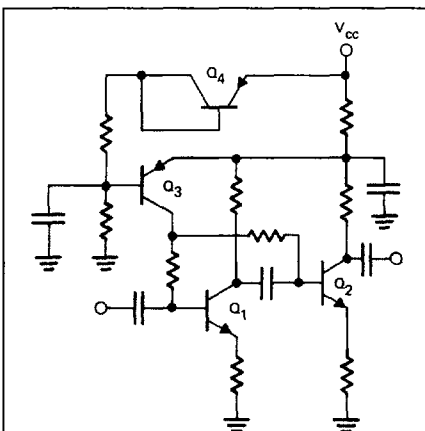


Fig. 14—Two-stage Active Bias.

extensive flexibility in achieving any desired gain, frequency noise figure, VSWR and power output.

The following sections give a few simple rules to use when cascading these amplifiers.

Gain

The typical cascaded gain will be the sum of the individual unit typical gains. The overall gain specification should not exceed the sum of the minimum gains as specified for each unit. The cascaded gain of the A15, A17 and PA10 is shown in Figure 15. An overall gain of 37 ± 0.4 dB is achieved from 10-1000 MHz.

Gain Flatness

Each cascable amplifier is flat across its respective band to within ± 0.3 dB. The cascaded gain flatness for up to 4 units typically will be less than ± 1.0 dB. However, for a conservative specification, ± 1.5 dB should be used when the full bandwidth is required.

The gain characteristics of three A5s in cascade are shown in Figure 16 for V_{CC} values of 15 V, 12 V, 10 V, and 8 V. Note that the gain flatness remains virtually unchanged for voltages ranging from 8 to 15 volts. The total power drain for the three

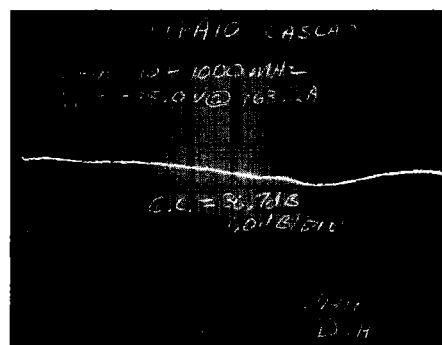


Fig. 15—Gain of the A15, A17 and PA10 Cascade.

units is only 288 mW at 8 volts compared to 1120 mW at 15 volts. This cascade is very useful in applications with limited power drain.

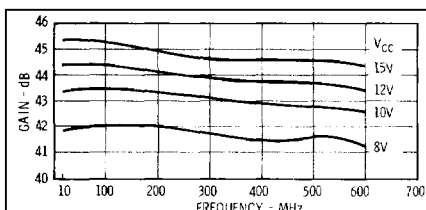


Fig. 16—Gain Performance of Three A5 Amplifiers in Cascade for V_{CC} Values of 15V, 12V, 10V and 8V.

Cascaded Noise Figure

The overall noise figure measured at the input amplifier of a cascade will increase due to the second stage contribution. The overall noise figure of a cascade can be calculated using the following equation:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \dots + \frac{F_N - 1}{G_2 \dots G_{N-1}}$$

where F_N is the noise figure of the Nth stage of G_N is the gain of the Nth stage. All terms are numeric ratios and are not in dB.

Table 1 gives the second stage contribution for the gains available from the cascable amplifiers. For example, a typical cascade would be an A1 followed by two A5s. If both A5s have a 5.0 dB noise figure, from Table 1, the contribution of the second A5 to the first A5 is only 0.093 dB. In this example, the second stage contribution to the overall noise figure is virtually the same even if the A1 (first stage Noise Figure of 3 dB) is cascaded with one or more A5s. The contribution given from Table 1 is 0.146 dB. If the second stage were an A1 (3 dB Noise Figure) instead of an A5, then the contribution would be 0.068 dB or an improvement of only 0.08 dB.

The disadvantages of cascading two or more low noise amplifiers are lower input power for 1 dB gain compression and larger gain variation over temperature because of the low bias level on the transistor. For example, the individual data sheet on the low-level A3 amplifier shows a gain variation of approximately ± 0.5 dB over a temperature range of -54°C to $+100^\circ\text{C}$. The variation for the A5 over the same temperature range is typically less than ± 0.3 dB. The main advantages of cascading low-level amplifiers are lower power drain and better limiting action to high-level signals.

Impedance

The Watkins-Johnson Company cascable amplifiers are designed for 50-ohm input and output impedance by using both collector-to-base and emitter resistive feedback. In addition, LC matching and reactive feedback compensation are used to obtain matching at the high end of the band. The photograph in Figure 17 shows the cascaded input and output impedance of an A11/A15 cascaded. In this case the cascaded VSWR is improved slightly over the unit VSWR. Most Watkins-Johnson cascable

Table 1—Second Stage Noise Figure Contribution—dB

First Stage Gain dB	Noise Figure dB	Second Stage Noise Figure						
		3 dB	5 dB	8 dB	10 dB	12 dB	15 dB	20 dB
20 dB	8.0	0.007	0.015	0.040	0.060	0.100	0.206	0.633
	5.0	0.014	0.03	0.070	0.120	0.200	0.401	1.183
	3.0	0.022	0.05	0.110	0.190	0.310	0.620	1.750
	1.5	0.039	0.07	0.160	0.270	0.430	0.852	2.307
15 dB	8.0	0.220	0.047	0.114	0.190	0.310	0.620	1.750
	5.0	0.043	0.093	0.225	0.370	0.600	1.160	2.989
	3.0	0.068	0.146	0.351	0.580	0.920	1.718	4.098
	1.5	0.096	0.205	0.488	0.800	1.250	2.267	5.074
10 dB	8.0	0.07	0.150	0.350	0.580	0.920	1.718	4.098
	5.0	0.14	0.290	0.670	1.090	1.670	2.941	6.160
	3.0	0.21	0.450	1.030	1.620	2.420	4.039	7.754
	1.5	0.30	0.620	1.390	2.140	3.120	5.008	9.036
7 dB	8.0	0.140	0.290	0.670	1.090	1.670	2.941	6.160
	5.0	0.270	0.560	1.260	1.950	2.870	4.672	8.601
	3.0	0.410	0.850	1.850	2.790	3.950	6.088	10.374
	1.5	0.570	1.160	2.430	3.560	4.910	7.264	11.756

amplifiers offer a bandwidth of 6-8 octaves. In general, the impedance is between 40-60 ohms and is essentially resistive below 1-1/2 octaves from the high end of the band.

VSWR

The VSWR on most of the amplifiers is specified at a maximum of 2.0:1. The cascaded VSWR will be typically less than 2.0:1 across the entire band, but can be greater than 2.0:1 in certain cases. About 1-1/2 octaves below the high end of the band there is virtually no degradation when cascading, and the 2.0:1 cascaded specification can be easily achieved. In general, when covering the entire bandwidth, the cascaded VSWR should be specified 2.5:1 maximum, with amplifiers that have a 2.0:1 specification.

Output Power

When cascading the cascable amplifiers, there is no degradation in output power of

the last stage provided that the driver stage provides sufficient power. Figure 18 shows the gain compression versus output power for various models. The curves extend from the uncompressed region of 2 dB of gain compression. The threshold compression point is defined as the 0.1-dB gain compression point and is the level of output power that the amplifier can deliver to the next stage without undergoing any significant loss of power output for the cascade.

The curves of Figure 18 show the additional compression that can be expected if the driving stage does not provide sufficient power. The necessary drive power is calculated by using the following equation:

$$P_D = P_O - G_O + 1$$

where P_O is the output power and G_O is the uncompressed gain of the stage being



Fig. 17—Input and Output Impedance of an A11/A15 Cascade.

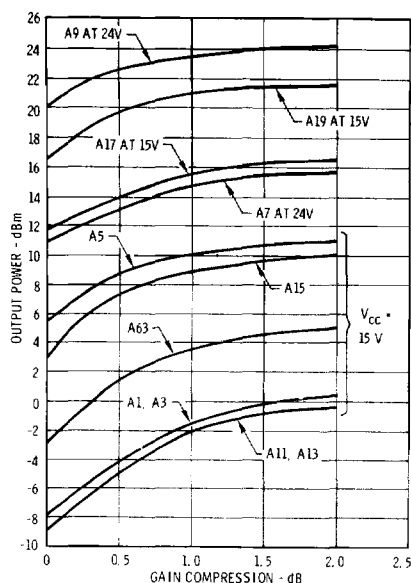


Fig. 18 - Gain Compression Versus Output

Table 2 - Typical Threshold Compression Point

Model	P _{Threshold} (0.1 dB Comp. Pt.)	P _{Driver} (0.1 dB Comp. Pt.)	Min. P _{Out} (1.0 dB Comp. Pt.)
PA10 & PA12	+19.0	+13.0	+22.5
A9	+21.0	+11.5	+22.0
A19	+17.0	+15.0	+21.0
A17	+12.5	+4.0	+15.0
A7	+11.5	0	+14.0
A5	+6.5	-6.0	+8.0
A15	+4.0	-7.0	+7.0
A63	-2.0	-12.0	+3.0
A1 & A3	-7.0	-15.0	-1.0
A11	-8.0	-16.0	-2.0
A79	+20.7	+9.4	+22.4
A82	+17.0	+0.2	+20.7
A87-2	+6.3	-4.7	+10.3
A86	+2.1	-19.1	+7.9
PA3	+21.1	+11.6	+25.1
A59	+20.6	+12.4	+22.4
RA89	+19.5	-2.3	+21.2
A88	+18.1	+2.4	+20.4
A58	+18.2	+9.2	+19.2
A72	+12.7	-0.7	+13.3
A74	+6.9	-19.4	+8.6
A83	-4.6	-29.8	-0.8
PA15	+22.4	+14.3	+26.8
A19-1	+20.9	+12.8	+23.3
A66-1	+12.1	-10.1	+15.9
A18-1	+11.7	+1.8	+15.8
A12	-0.7	-6.5	+8.5
A39	+20.8	+16.4	+22.9
A38	+18.8	+14.3	+20.8
A36-1	+6.1	+0.4	+15.4

driven. The minimum P_D is the threshold compression point as previously defined.

Typical threshold compression points for several Watkins-Johnson Company cascable amplifiers are given in the first column of Table 2. The center columns show the threshold driving power required to obtain the output power listed in the right hand column. This table is generated by using the above equation. Table 2 shows that a P_D of +11.5 dBm is required to drive the A9 to a P_O of +22 dBm output power. The A7 provides this amount of power at its threshold compression point. The A19 requires +15 dBm. The A17 provides +15 dBm at its 1 dB compression point; hence, an output power reduction of 1 dB would typically result when cascading the A17 and A19. The A17 requires +4.0 dBm to obtain a +15 dBm output which is provided by the A15. If lower noise is needed, then the A63 should be used, which is compressed about 1 dB at the +4 dBm output level. Thus, the overall result when cascading the A63, A17 and A19 is approximately a 2 dB loss in output power. The measured output power of this cascade is shown in Figure 19. The output powers shown in Figure 18 were measured at 100 MHz. Generally, the output power increases as the high end of the band is approached because of the reactive matching effect.

The output power of a single stage is measured using a 50-ohm source and a 50-ohm load. The output power of the driving

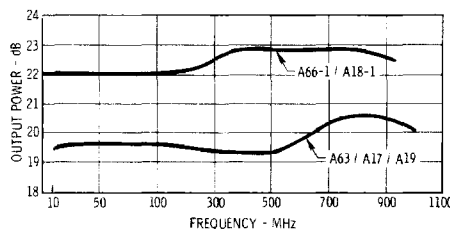


Fig. 19—Power Output of the A66-1, A18-1, A63, A17 and A19 Cascade.

amplifier can be affected by the load (input impedance of next stage) presented to it. This can cause some minor differences to occur that are not fully explained by considering compression alone. In general, if the effective load presented to the driver stage is greater than 50 ohms, then more output power can be obtained from the driver stage. Conversely, if the load presented is less than 50 ohms, then a reduction may result.

The WJ-A9 can be driven to provide 200-300 mW (+25 dBm) of output power with additional compression, and the A19 can be driven to provide 150-200 mW (+23 dBm) maximum. Beyond these limits, the bias current of the amplifiers will increase because the RF signal effect on the base bias tends to cause Class C operation. Because the gain of the A19 is lower than the A9, a larger RF signal will be present at the base, which causes the base bias to be affected at

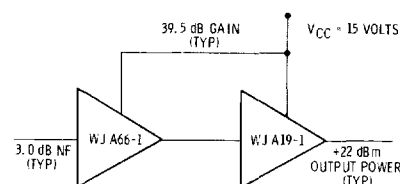


Fig. 19A

a lower output power than on the A9. For maximum reliability it is not recommended that the A9 or A19 be operated above these output power levels.

The A63/A17/A19 was introduced in late 1974 and has remained a state-of-the-art cascade for several years. Now there is a cascaded combination providing equal or superior results using only two TO-8 amplifiers as shown in Figure 19A.

The A66-1/A19-1 cascade provides about 3 dBm higher output power, equal NF to

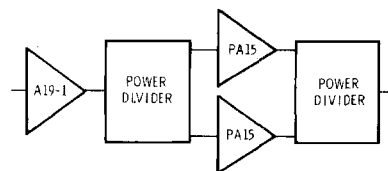


Fig. 20 - Higher Output Power by Paralleled Amplifier Operation.

the WJ-A63, and 4 dB higher gain. The primary reason for the higher output of this cascade is simply that the A19-1 requires about 4 dB less drive power (see gain compression curves in Figure 18) than the A19 because of its higher gain.

There are now units within the Watkins-Johnson product line of cascable amplifier models that more than double the output power of the industry pace-setting A9 and A19 TO-8 amplifier. The WJ-PA3 will provide a minimum of 200 milliwatts at 15 volts and, in addition, offer 3.5 dB higher gain than the WJ-A9 amplifier.

For applications out to 1000 MHz, the WJ-PA10 will provide a minimum of 200 milliwatts of output power (@ 20V V_{CC})

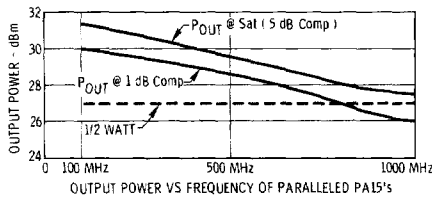


Fig. 20A

with a 3.5 dB more gain than the WJ-A19 amplifier. And now there is the WJ-PA15 TO-8 amplifier which provides 3.0 dB even higher gain (13.5 dB typically) out to 1000 MHz and 200 milliwatts minimum output power using only a 15 volt V_{CC} supply. In addition, the WJ-PA15 provides a typical output power of 0.5 watt out to 500 MHz.

When the output power required is higher than that which is provided by a single output stage, then the units can be paralleled using reactive power dividers as shown in Figure 20. Nearly 3 dB of additional output power can be obtained using this method. An example of paralleled output (saturated) power is shown in Fig. 20A.

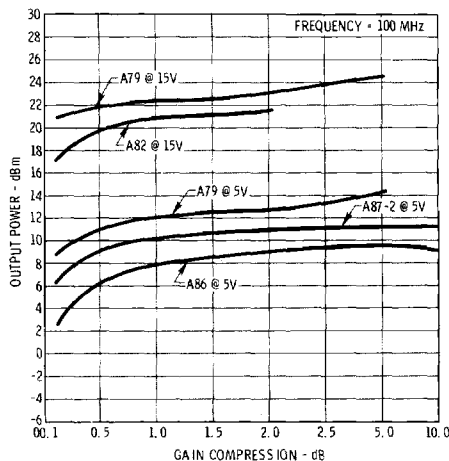


Fig. 21-A

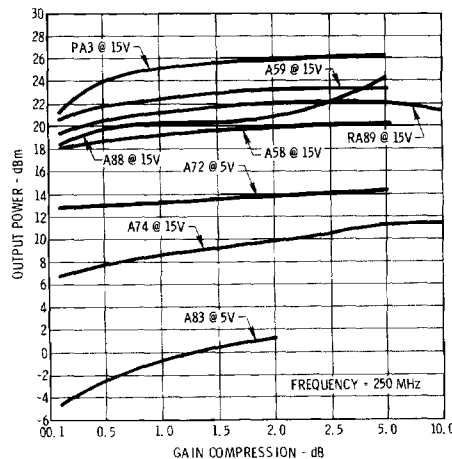


Fig. 21B

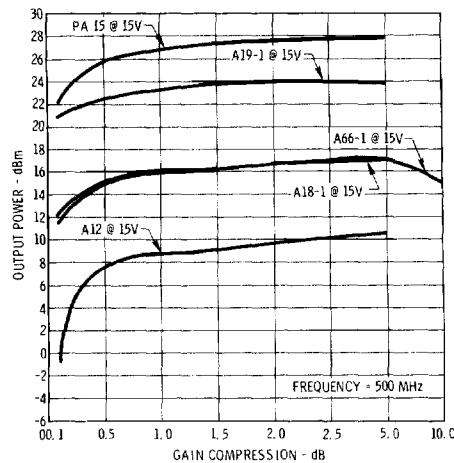


Fig. 21C

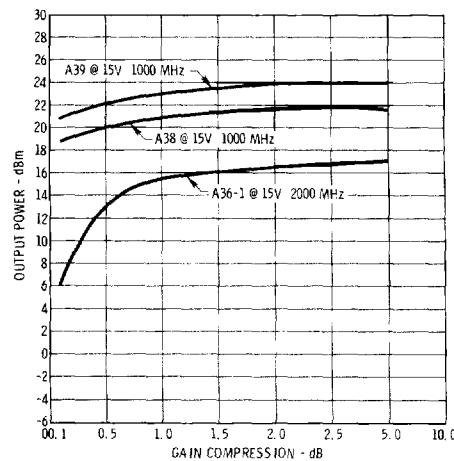


Fig. 21D

When the paralleled pair of PA15s is driven by a WJ-A19-1 or WJ-PA10 amplifier, there is virtually no change in the output power.

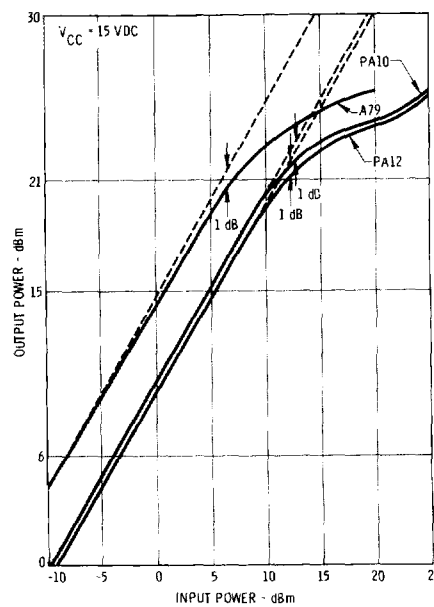


Fig. 22A

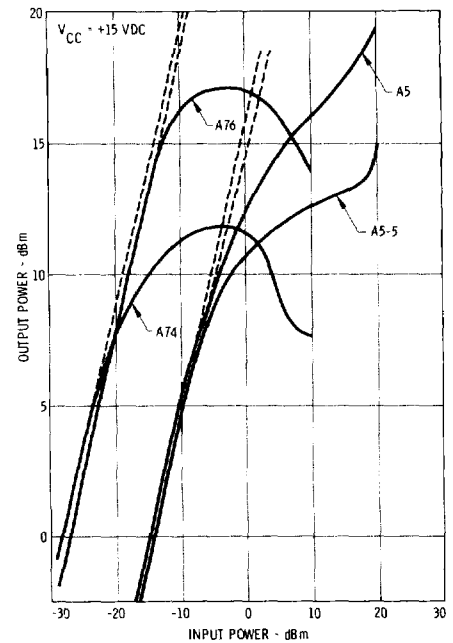


Fig. 22B

Saturated Output Power

Several new amplifier gain-compression versus output-power curves have been added in Figures 21A through 21D. These curves show the gain compression characteristic for many newer-generation TO-8 amplifiers for input levels that cause the onset of compression to moderate levels of saturation where the gain compression is 5 to 10 dB.

These curves demonstrate a relatively monotonic characteristic of gain compression for moderate saturation levels.

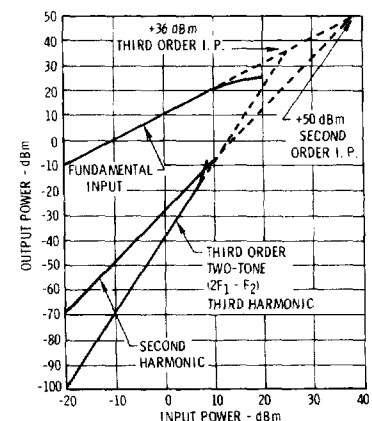


Fig. 23—Output Power Versus Input Power of the WJ-A9 Amplifier.

What happens when the W-J Cascadable Amplifiers are driven into heavy saturation depends on the bias design of the circuit and whether or not the units are single-stage or two-stage. Figures 22A and 22B give some exemplary curves of overdrive conditions.

Distortion Products and Intercept Point

Due to the inherent nonlinearities of the amplifiers, intermodulation and harmonic distortion products are generated that limit the dynamic range. The level of the distortion product relative to the signal level can be calculated from the intercept point, or conversely, the intercept point can be calculated from the relative suppression. The intercept point (I.P.) is the theoretical point of intersection of the fundamental slope with the slope of the distortion product, and is shown in Figure 23 for the WJ-A9 amplifier. Third-order two-tone distortion is the mixing of two signals applied at the input of the amplifier. These third-order products will appear at the frequencies $2f_1 - f_2$, $2f_1 + f_2$, $2f_2 + f_1$ and $2f_2 - f_1$. Two-tone second-order products will appear at the frequencies $f_1 + f_2$ and $f_1 - f_2$. The intercept point can be calculated from equation 1 :

$$1) \text{ I.P.} = \frac{R_{dB}}{n-1} + S \text{ dBm}$$

where R is the relative suppression below the fundamental in dB, n is the order number and S is the absolute signal level in dBm. If the intercept point is known, the relative suppression can be calculated by equation 2:

$$2) R_{dB} = (n-1) (\text{I.P.} - S)$$

The intercept point will be the output intercept point if S is the output signal level and it will be the input intercept point if S is the input signal level. For example, what is the typical third-order two-tone intercept point for the A9? From Figure 23, the value of R is 52 dB for an output power of +10 dBm. Using equation 1 :

$$\text{I.P.} = \frac{52}{3-1} + 10 = +36 \text{ dBm}$$

Table 3 shows a summary of the second-order harmonic, third-order harmonic and third-order, two-tone intercept point for several amplifiers. The second-order two-

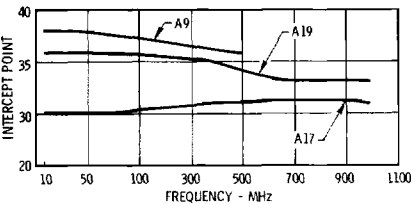


Fig. 24—A9, A19 and A17 Output Third-Order

tone intercept points are not shown since they are approximately equal to the second-order harmonic intercept points.

The A9, A19, and A17 output third order two-tone intercept points versus frequency are shown in Figure 24. Note how the A17 I.P. approaches the A19 I.P. at the high end of the band. When cascading amplifiers, if the driving amplifier output intercept point is equal to the input intercept point of the amplifier being driven, then each amplifier would contribute equally, causing a 3-dB reduction in the output intercept point of the output stage. If there is 6 dB of margin on the driver stage, then approximately 1 dB of degradation could result. As a rough rule of thumb, the output intercept point will be typically 12 dB higher than the output power on the Watkins-Johnson Company cascadable amplifiers. Any degradation in output power will generally cause degradation in the intercept point.

Intercept Degradation

Distortion in RF Signal Processing is often a baffling problem for the designer. Basic understanding of what causes distortion, and how it propagates within the system will help optimize performance. Intermodulation distortion is of key interest in a newly developed system.

The two-tone intercept point is often used as a measure of intermodulation distortion in signal processing devices. When amplifiers are cascaded the degradation of the intercept point can be easily determined.

When a signal is amplified, the output is characterized by the primary frequencies, f_1 and f_2 , and their intermodulation products (Table 1). To specify the level of distortion due to the intermodulation products, the intercept point definition was created which, simply stated, defines the nth order output intercept point as,

$$(1) \quad n_{IP} = \frac{n_R}{(n-1)} + P_O$$

n_R is the relative suppression of the nth order product in dB and P_O is the total power output of the amplifier in dBm.

When two amplifiers are cascaded, the cascade intercept point is a function of the intercepts of the individual amplifiers and

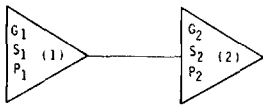


Fig. 25

the gain of the system. To understand the degradation of intercept point, the relative suppression of intermodulation products must be understood.

Relative suppression is the ratio of the total output power (mw) to the power level of an intermodulation spur. This is valid for all orders.

$$(2) \quad n_r = \frac{P_O}{n_S}$$

In a two-amplifier system (see Figure 25), if the individual suppression characteristics are known, the total suppression can be determined.

Both amplifiers generate intermodulation products, n_{S1} and n_{S2} respectively. The first product, n_{S1} is amplified by the gain of the second amplifier and then combines with n_{S2} . If it is assumed that all input and output impedances match, then the intermodulation products' sum is,

(3)

The gain of the second amplifier is represented by g_2 . With some mathematical manipulation, a formula for the total relative suppression n_{rt} can be obtained:

$$(3a) \quad \frac{\sqrt{S_T^n}}{\sqrt{P_O}} = \frac{\sqrt{S_1^n \times g_2}}{\sqrt{P_O}} + \frac{\sqrt{S_2^n}}{\sqrt{P_O}}$$

$$(3b) \quad \frac{\sqrt{S_T^n}}{\sqrt{P_O}} = \frac{\sqrt{S_1^n \times g_2}}{\sqrt{P_1 \times g_2}} + \frac{\sqrt{S_2^n}}{\sqrt{P_O}}$$

$$(4) \quad \frac{1}{\sqrt{r_T^n}} = \frac{1}{\sqrt{r_1^n}} + \frac{1}{\sqrt{r_2^n}}$$

$$(4a) \quad r_T^{n-1/2} = r_1^{n-1/2} + r_2^{n-1/2}$$

Table 3 - Second-Order Harmonic, Third-Order Harmonic and Third-Order, Two-Tone Intercept Points for Several Amplifiers

Model	Second-Order Harmonic	Third-Order Harmonic	Third-Order Two-Tone
	I.P.	I.P.	I.P.
A5	+40 dBm	+27 dBm	+22 dBm
A7	+45 dBm	+30 dBm	+26 dBm
A9	+50 dBm	+36 dBm	+36 dBm
A11	+15 dBm	+13 dBm	+10 dBm
A15	+35 dBm	+24 dBm	+20 dBm
A17	+45 dBm	+32 dBm	+29 dBm
A19	+46 dBm	+34 dBm	+34 dBm

If the cascaded relative suppression is known, the degraded intercept point can be calculated:

$$(4b) \left(r_1^n \right) \left(\frac{1}{n-1} \right) \frac{-(n-1)}{2} =$$

$$\left(r_1^n \right) \left(\frac{1}{n-1} \right) \frac{-(n-1)}{2} +$$

$$\left(r_2^n \right) \left(\frac{1}{n-1} \right) \frac{-(n-1)}{2}$$

$$(4c) \left[r_1^n \left(\frac{1}{n-1} \right) P_o \right] \frac{-(n-1)}{2} =$$

$$\left[\left(r_1^n \left(\frac{1}{n-1} \right) \frac{-(n-1)}{2} + \right. \right.$$

$$\left. \left(r_2^n \left(\frac{1}{n-1} \right) \frac{-(n-1)}{2} \right) \right] P_o \frac{-(n-1)}{2}$$

$$(4d) \left[r_1^n \left(\frac{1}{n-1} \right) P_o \right] \frac{-(n-1)}{2} =$$

$$\left[\left(r_1^n \left(\frac{1}{n-1} \right) P_o \right) \frac{-(n-1)}{2} + \right.$$

$$\left. \left(r_2^n \left(\frac{1}{n-1} \right) P_o \right) \frac{-(n-1)}{2} \right]$$

$$(5) \left[r_1^n \left(\frac{1}{n-1} \right) P_o \right] \frac{-(n-1)}{2} =$$

$$\left(r_1^n \left(\frac{1}{n-1} \right) P_1 g_2 \right) \frac{-(n-1)}{2} +$$

$$\left(r_2^n \left(\frac{1}{n-1} \right) P_2 \right) \frac{-(n-1)}{2}$$

If equation (1) is converted from dBm to mW, it can be substituted into (5) and a formula for cascaded intercept point can be obtained:

$$(1) IP^n = \frac{R^n}{n-1} + P_o$$

$$(1a) IP^n = n_r \left(\frac{1}{n-1} \right) P_o$$

Rewriting (5) using (1a):

$$(6) IP_T^n \frac{-(n-1)}{2} = \left(IP_1^n g_2 \right) \frac{-(n-1)}{2} +$$

$$\left(IP_2^n \right) \frac{-(n-1)}{2}$$

This can be generalized to the case of x amplifiers in cascade:

$$(7) IP_T^n \frac{-(n-1)}{2} =$$

$$\sum_{i=1}^x \left(IP_i^n g_{(i+1,x)} \right) \frac{-(n-1)}{2}$$

The gain term represents the gain following the i th amplifier. This formula is valid for all orders. However, it gives worst case

performance, assuming a perfect 50Ω system. If a severe mismatch occurs in the cascade, further degradation can result.

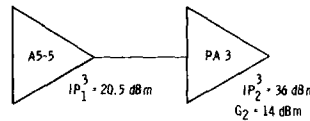


Fig. 26

This formula can be easily validated experimentally. Looking at the Watkins-Johnson A5-5 in cascade with the PA3 the intercept point at 33.0 dBm can be measured using the individual amplifier characteristics (see Figure 26) and the cascade formula, it can be seen that the 3 dB degradation, in third order intercept point is to be expected:

$$IP_1^3 = 20.5 \text{ dBm} = 112 \text{ mW}$$

$$IP_2^3 = 36.0 \text{ dBm} = 3981 \text{ mW}$$

$$g_2 = 14.7 \text{ dB} = 29.5$$

$$IP_T^3 = [(112 \text{ mW}) (29.5)]^{-1} + [3981 \text{ mW}]^{-1}$$

$$IP_T^3 = 1808 \text{ mW} = 32.6 \text{ dBm}$$

The intercept formula can be used to introduce several rules of thumb for

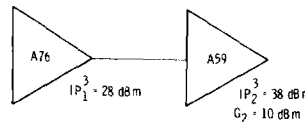


Fig. 27

$$IP_1^3 = 28 \text{ dBm} = 630.0 \text{ mW}$$

$$IP_2^3 = 38 \text{ dBm} = 6309 \text{ mW}$$

$$g_2 = 10 \text{ dB} = 10$$

$$IP_T^3 = [(630.9 \text{ mW}) (10)]^{-1} + [6309 \text{ mW}]^{-1}$$

$$IP_T^3 = 3154 \text{ mW} = 35.0 \text{ dBm}$$

$$\Delta ip = 3 \text{ dB}$$

cascading amplifiers. The first rule applies (see Figure 27) when the output intercept point of the first amplifier equals the input intercept point of the second. Each amplifier will contribute equally to the intermodulation product, causing a 3 dB degradation in total output intercept point.

The second rule is demonstrated when the output intercept point of the first amplifier is much greater than the input intercept point of the second amplifier. (See Figure 28.)

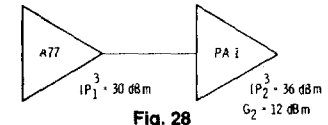


Fig. 28

$$IP_1^3 = 30 \text{ dBm} = 1000 \text{ mW}$$

$$IP_2^3 = 36 \text{ dBm} = 3981 \text{ mW}$$

$$g_2 = 12 \text{ dB} = 15.8$$

$$IP_T^3 = [(1000 \text{ mW}) (15.8)]^{-1} + [3981 \text{ mW}]^{-1}$$

$$IP_T^3 = 3182 \text{ mW} = 35.0 \text{ dBm}$$

$$\Delta ip = 1 \text{ dB}$$

In this case a 6-dB overlay causes a 1-dB degradation in intercept point. The third rule of thumb is that a mismatch of intercept point will cause degradation equal to or greater than the mismatch itself. In the last example, the output intercept point of the first amplifier is 6 dB less than the input intercept point of the second. (See Figure 29.)

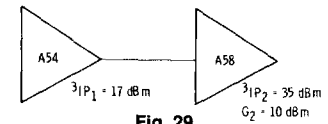


Fig. 29

$$IP_1^3 = 19 \text{ dBm} = 79.4 \text{ mW}$$

$$IP_2^3 = 35 \text{ dBm} = 3162 \text{ mW}$$

$$g_2 = 10 \text{ dB} = 10$$

$$IP_T^3 = [(79.4 \text{ mW}) (10)]^{-1} + [3162 \text{ mW}]^{-1}$$

$$IP_T^3 = 634.8 \text{ mW} = 28.0 \text{ dBm}$$

$$\Delta ip = 7 \text{ dBm}$$

Intermodulation distortion can be reduced through careful design. The general cascade formula can be used to calculate degradation in intercept point. For quick estimation in cascading, the following three rules of thumb can be used:

- 1) If the output intercept point equals the input intercept point, a 3-dB degradation will result.
- 2) If the output intercept point is much greater (> 6 dB) than the input, IP less than 1 dB degradation can be expected.
- 3) If the output IP is much less (< 6 dB) than the input IP, degradation equal to the mismatch, or more, can be expected.

Through the use of the three rules and the general cascade formula, intermodulation distortion can be minimized.

Intercept Degradation and Dynamic Range

The dynamic range (DR) of cascable amplifiers for a single input signal can be defined as: the difference between the input signal level that causes 1 dB of gain compression and the minimum input signal level that can be detected above the amplifier's noise level. This minimum detectable signal (defined as 3 dB above the noise level) MDS, can be expressed in dBm by the equation:

$$\text{MDS} = -111 \text{ dB/ MHz} + B + \text{NF}$$

where B is the bandwidth relative to 1 MHz in dB, and NF is the noise figure measured at the amplifier input. The input

signal or drive power (P_D) that produces 1 dB of gain compression is given by the equation:

$$P_D = P_{\text{output}} - \text{Gain} + 1 \text{ dB}$$

where P_{output} is the output power at the 1 dB compression point. The dynamic range, therefore, is the difference between P_D and MDS:

$$\text{DR} = P_D - \text{MDS}$$

Using the data given in Figure 15 (A63, A17, A19) and a 1 MHz bandwidth, DR is calculated as:

$$\begin{aligned} \text{DR} &= (P_{\text{output}} - \text{Gain} + 1 \text{ dB}) - (-111 \\ &\quad \text{dBm/ MHz} + B + \text{NF}) \\ \text{DR} &= (+19 - 35 + 1) - (-111 + 0 + 4) \\ \text{DR} &= 92 \text{ dB (1 MHz bandwidth)} \end{aligned}$$

For a 500 MHz bandwidth there would be a 27 dB reduction in the overall dynamic range for the above example.

$$(10 \log \frac{500}{1} = 27 \text{ dB})$$

Another definition of dynamic range is the "spurious-free" region which characterizes the effective performance of an amplifier with more than one signal applied to the input. For the case of input signals at equal level, the spurious free dynamic range, $\text{DR}_{\text{spur-free}}$ is given by the equation:

$$\text{DR}_{\text{spur-free}} = 2/3 (\text{I.P.} - \text{Gain} - \text{MDS})$$

where I.P. is the third order two-tone intercept point in dBm. Using the same example and an I.P. of +32 dBm for a A19 amplifier, the spurious-free dynamic range is:

$$\begin{aligned} \text{DR}_{\text{spur-free}} &= 2/3 (+32 - 35 - (-111 \\ &\quad + 0 + 4.0)) \\ \text{DR}_{\text{spur-free}} &= 69 \text{ dB (1 MHz bandwidth)} \end{aligned}$$

Phase Linearity and Group Delay

The Watkins-Johnson Company cascable amplifiers offer excellent linearity and relatively constant group delay because of thin-film integrated construction. Phase linearity is the best fit straight line to the phase response curve and is expressed in degrees deviation. Computer printouts of phase linearities from an automatic network analysis system are shown in Tables 4 and 5. The first column shows the deviation from linear phase and the last column shows the group delay.

The phase linearity of the WJ-A5 is typically less than $\pm 2^\circ$ across a 10-500 MHz band. The phase linearity of WJ-A15 is typically $< \pm 6^\circ$ across its entire 5-1000 MHz band and the WJ-A25 is typically less than $\pm 7^\circ$ across its entire 5-1500 MHz band. Substantial improvement in phase linearity can be obtained by not using the upper end of the band. Table 4 compares the phase linearity of the A15 across the 110-510 MHz and 110-1010 MHz bands. The linearity across the 110-510 MHz band is typically within $\pm 0.5^\circ$. The A25 phase linearity is typically $\pm 1.8^\circ$ up to 1000 MHz as shown in Table 5.

The cascaded phase linearity will be equal to or less than the sum of the individual stage linearity. Thus, two stages of A15s would typically have $\pm 1.0^\circ$ phase linearity

TABLE 4

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
100.	.24	.00	.12	14.91	.50
200.	-.37	-18.07	.11	14.90	.49
300.	-.10	-35.28	.11	14.89	.47
400.	.40	-52.21	.07	14.85	.49
500.	-.14	-70.20	-.12	14.67	.50

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
100.	-3.78	.00	.19	14.92	.50
200.	-2.40	-18.06	.14	14.87	.49
300.	-.30	-35.41	.16	14.89	.48
400.	2.01	-52.54	.08	14.81	.49
500.	3.42	-70.57	-.09	14.65	.52
600.	3.60	-89.83	-.16	14.58	.55
700.	3.08	-109.79	-.09	14.64	.56
800.	2.06	-130.26	-.05	14.68	.61
900.	-1.61	-153.37	-.05	14.69	.65
1000.	-6.08	-177.28	-.19	14.55	.66

TABLE 5

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
100.	-1.04	.00	.14	10.58	.37
200.	-.75	-13.24	.06	10.50	.37
300.	-.43	-26.45	.14	10.58	.36
400.	.51	-39.03	.09	10.53	.35
500.	1.70	-51.37	.05	10.49	.37
600.	1.20	-65.37	-.10	10.34	.39
700.	.51	-79.61	-.13	10.32	.39
800.	.01	-93.34	-.10	10.35	.39
900.	-.23	-107.40	-.14	10.30	.41
1000.	-1.80	-122.50	-.12	10.32	.42

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
100.	-4.02	.00	.16	10.61	.37
200.	-2.90	-13.22	.08	10.53	.37
300.	-1.86	-26.52	.22	10.66	.36
400.	-.17	-39.16	.10	10.54	.35
500.	1.68	-51.65	.01	10.46	.37
600.	2.12	-65.55	-.11	10.34	.39
700.	2.29	-79.72	-.14	10.31	.39
800.	2.92	-93.42	-.13	10.32	.39
900.	3.20	-107.48	-.17	10.28	.40
1000.	2.49	-122.53	-.15	10.29	.41
1100.	2.14	-137.21	-.03	10.42	.41
1200.	1.89	-151.00	-.04	10.41	.42
1300.	.37	-167.65	-.05	10.39	.47
1400.	-3.05	-185.41	-.11	10.33	.50
1500.	-7.09	-203.79	-.22	10.23	.51

TABLE 6

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
500.	.24	.00	.28	16.55	.54
600.	-.48	-19.40	.36	16.63	.52
700.	.16	-37.45	.33	16.60	.50
800.	.97	-55.32	.21	16.48	.52
900.	.30	-74.68	-.01	16.26	.55
1000.	-.91	-94.57	-.17	16.10	.53
1100.	-.41	-112.75	-.25	16.02	.51
1200.	-.21	-131.23	-.36	15.91	.52
1300.	-.70	-150.41	-.25	16.02	.51
1400.	.22	-168.16	-.19	16.08	.50
1500.	.81	-186.26	-.23	16.04	.50

FREQ. MHz	DEV LIN O DEG	REL O DEG	GAIN DEV dB	ABS GAIN dB	GROUP DELAY N—SEC
100.	2.09	.00	.28	16.58	.61
200.	-.74	-21.83	.19	16.49	.58
300.	-1.31	-41.41	.18	16.48	.54
400.	-1.62	-60.72	.26	16.55	.52
500.	-.89	-79.00	.23	16.52	.51
600.	-.51	-97.62	.33	16.63	.51
700.	.15	-115.96	.32	16.61	.51
800.	.78	-134.34	.20	16.50	.52
900.	.64	-153.48	-.00	16.30	.55
1000.	-.48	-173.60	-.15	16.15	.54
1100.	-.06	-192.19	-.19	16.11	.52
1200.	.11	-211.02	-.00	16.00	.53
1300.	.06	-230.08	-.29	16.00	.51
1400.	1.18	-247.96	-.23	16.07	.50
1500.	2.43	-265.71	-.28	16.01	.52
1600.	1.80	-285.34	-.33	15.97	.55
1700.	.93	-305.22	-.27	16.03	.55
1800.	.07	-325.08	-.16	16.14	.56
1900.	-1.01	-345.17	-.21	16.09	.58
2000.	-3.59	-366.75	-.06	16.24	.60

up to 500 MHz, but would degrade to about $\pm 12^\circ$ up to 1000 MHz.

The phase linearity of the A34 is shown in Table 6. The A34 is a two-stage 100-2000 MHz unit which offers typically $\pm 3.5^\circ$ across its entire bandwidth. Also shown is the phase linearity for the 500-1500 MHz band for which the A34 yields typically $\pm 1.0^\circ$

The group delay is defined as the rate of change of phase shift versus frequency

$$\frac{d\theta}{d\omega (\text{omega})}$$

The group delay is calculated by taking the phase difference over a given frequency band and then dividing by that frequency band. Both the phase and frequency are in radians. The group delay can be expressed in units of time by the mathematical relationship

$$t_d (\text{seconds}) = \frac{\text{Phase (degrees)}}{\text{frequency (Hz)} \times 360^\circ}$$

The group delay for the A25 amplifier is typically less than 0.5 nanoseconds up to 1000 MHz as shown from Table 5.

The excellent phase linearity and constant group delay properties of the A25 make it an ideal choice for Gigabit Data Rate 452

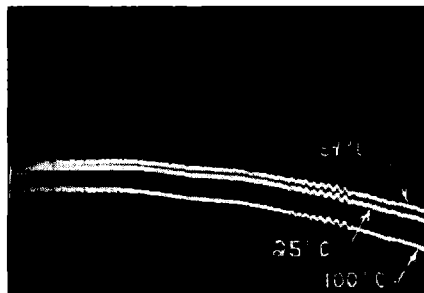


Fig. 30—Phase Shift of the WJ-A15 Amplifier.

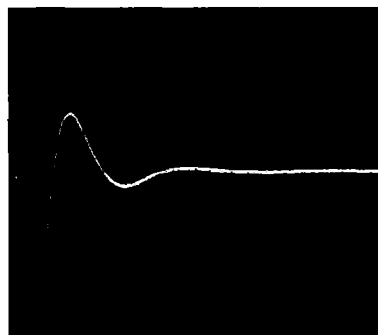


Fig. 31—Single A5.

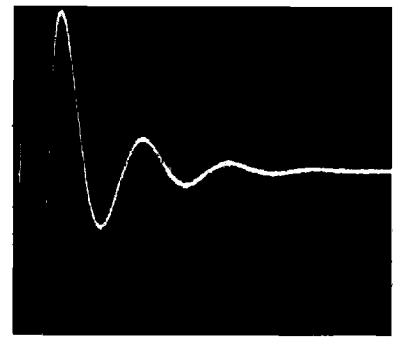


Fig. 32—Two Cascaded A5s.

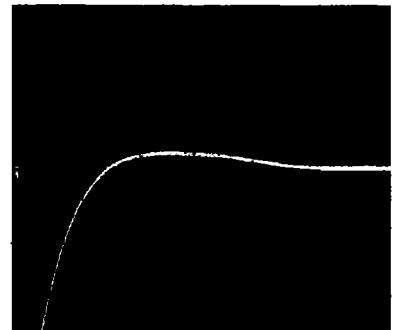


Fig. 33—Single A15.

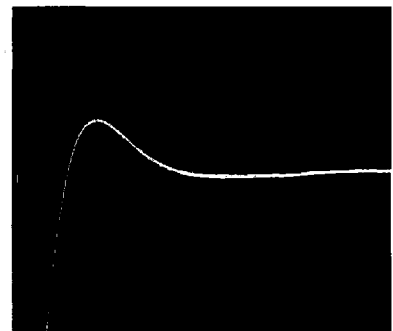


Fig. 34—Two Cascaded A15s.

Table 7—Amplifier Recovery Time
(To a 1 mV Setting Level)
After a 50 nsec Input Pulse is Applied

Model	0.1 Volt Input	1.0 Volt Input
A3	0.8 μ sec	1.0 μ sec
A5, A7	2.0 μ sec	2.5 μ sec
A9	1.3 μ sec	1.5 μ sec
A1, A11, A11-1, A15	3.0 μ sec	3.5 μ sec
A17	2.5 μ sec	3.5 μ sec
A19	0.8 μ sec	1.5 μ sec
A25	2.0 μ sec	3.5 μ sec
G1		1.5 μ sec

Systems. A convenient carrier frequency for the quadriphase modulated phase shift keying (QPSK) is 1 GHz, where the A25 provides high phase linearity and constant group delay.

Figure 30 shows the phase shift of a WJ-A15 amplifier operating at 1) room temperature, 2) at -54°C and 3) at $+100^\circ\text{C}$. There is virtually no change in linearity over this temperature range. Phase tracking is typically within $\pm 2^\circ$ between units.

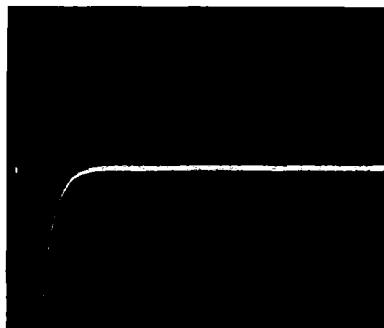


Fig. 35—Single A3.
Scales Common to Figures 9 to 14:
Vertical = 0.1 Volt/Div.
Horizontal = 0.5 μ Sec/Div.

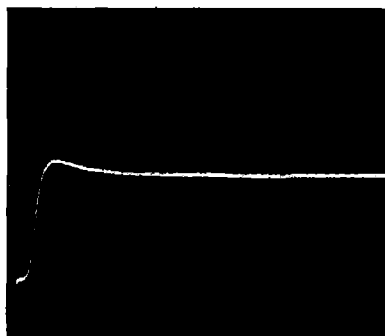


Fig. 36—Two Cascaded A3s.
Input Pulse:
Magnitude = 1.0 Volt
Width = 50 nsec.
Risetime = ~ 4nSec.



Fig. 37—Two Cascaded A5s with 50 pF Series Capacitor on Input.

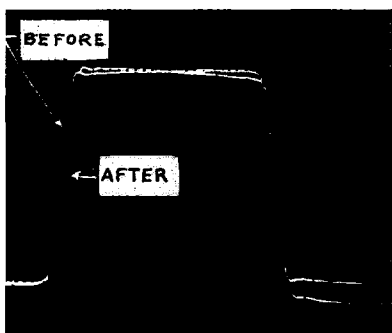


Fig. 38—Single A25.

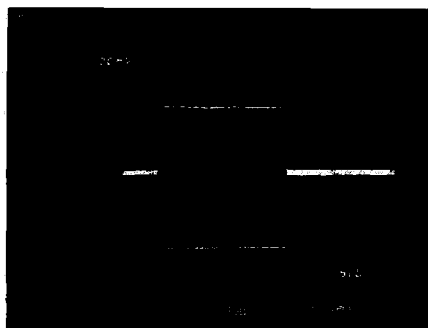


Fig. 39—A63 with 50 pF Series Capacitors on Input and Output at 1 dB Compression.

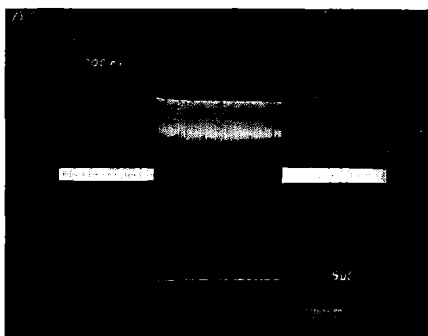


Fig. 39A—A63 with 50 pF Series Capacitors on Input and Output in Heavy Saturation.

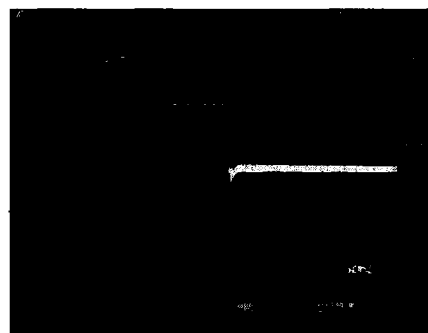


Fig. 40—A63 with 50 pF Series Capacitors on Input and Output at 1 dB Compression.

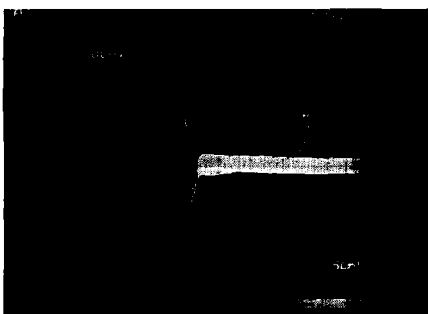


Fig. 40A—A63 with 50 pF Series Capacitors on Input and Output in Heavy Saturation.

Pulse Response

The recovery time of the A5, A15 and A3 amplifiers to a 1 volt, 50 μ sec video pulse are shown in Figures 31, 33 and 35 respectively; each amplifier consists of a single stage. Figures 31, 34 and 36 show the recovery time when the amplifier consists of two stages.

Each amplifier has a circuit feature that is different from the other two. Figures 31 and 32 represent the RF choke decoupling from the power supply (in addition to an RC network). Figures 33 and 34 represent active bias, and Figures 35 and 36 represent the resistive decoupling and passive emitter bias. The recovery time of the A5 is faster than the A15. The A3 yielded the lowest overshoot and the fastest recovery time. Table 7 shows the typical recovery time of each amplifier for various input pulse levels.

The overshoot and recovery time of the two cascaded A5s, shown in Figure 32, are reduced by adding a 50 pF capacitance in series with the input signal, and is shown in Figure 37. The recovery time is decreased from 3.5 μ sec to approximately 1 μ sec (to a 10 mV level). The response of the WJ-A25 to a 50 nsec, 0.1 video pulse is shown in Figure 38. The pulse is shown both before and after passing through the A25. The pulse droop is only 5%, indicating an excellent low frequency response down to approximately 1 MHz. Both the input pulse and the output pulse exhibit a rise time of approximately 4.5 nanoseconds, thus demonstrating the ultra-wide bandwidth property of the A25 amplifier.

A CW pulse will recover faster than a video pulse, as shown by Figure 39. Here a 500 MHz 20 μ sec wide CW pulse is shown after passing through the WJ-A63. The input power level was sufficient to cause approximately 1 dB of compression. The input and output lines are AC coupled



Fig. 41—WJ-A63 Temperature Performance.

with 50 pf capacitors which yield a recovery time as shown by Figure 40 of approximately 20 nsec. Without the capacitors the recovery time (not shown) is 250-300 nsec.

When the unit is driven with a +5 dBm input signal, which is sufficient to cause heavy saturation, the recovery time increases to only approximately 25 nsec as shown by Figure 40A. When the 50 pf capacitors are not used, severe pulse distortion occurs and the recovery time in

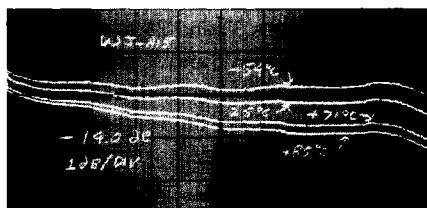


Fig. 42—WJ-A15 Temperature Performance.

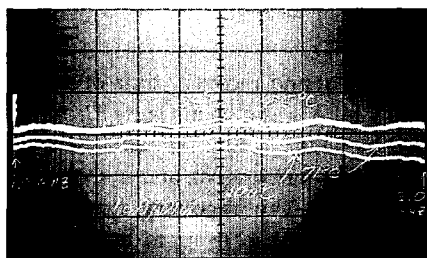


Fig. 43—WJ-A33 Temperature Performance.

the saturation mode extends to approximately 5 μsec.

Temperature

Each Watkins-Johnson Company cascable amplifier has been designed for optimum performance over a full temperature range. Typical gain changes over ambient temperature ranges of -54°C to +85°C are less than ±0.3 dB for the 5-500 MHz units, and less than ±1.0 dB for the higher frequency units. The actual

Table 8—Actual Amplifier Case Temperature When the Ambient Temperature is at 25°C

T Model	Case	V _{CC}	Dissipation Watts	Thermal Impedance Case to Ambient with Circuit Board (°C/W)	Approx. Area of Circuit Board
A5	32°C	15V	0.37	19°C/W	10 in ²
A7	42°C	24V	1.00	17°C/W	10 in ²
A17	36°C	15V	0.66	17°C/W	10 in ²
A19	49°C	15V	1.5	16°C/W	10 in ²
A9	34°C	24V	2.60	3.5°C/W	40 in ²

temperature performance of an A63 amplifier over a 10 to 1000 MHz bandwidth is shown in Figure 41.

The A63 amplifier gain is typically 16 dB across its full bandwidth at 25°C, with less than 0.8 dB degradation at 85°C. The temperature performance of the WJ-A15 is shown in Figure 42. Again, there is only about a 0.8 dB loss of gain at 85°C. The temperatures are ambient with the 85°C temperature line representing a unit's case temperature of about 95°C on the A15. Figure 43 shows the WJ-A33 TO-8 amplifier temperature performance across the 1.0 to 2.0 GHz frequency range. Note at 100°C and 2 GHz the gain changed by only 0.9 dB, and by only 0.4 dB at 71°C. These photographs display the excellent temperature performance engineered into the Watkins-Johnson Company cascable amplifiers.

Table 8 shows some actual amplifier case temperatures when the ambient temperature is at 25°C. In a typical application, the units are mounted with the case making good contact to the ground plane of the circuit board.

Because the A7, A9, and A19 dissipate 1 watt or more of power, 20 in² of area

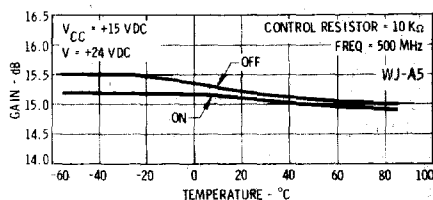


Fig. 44—Gain vs. Temperature.

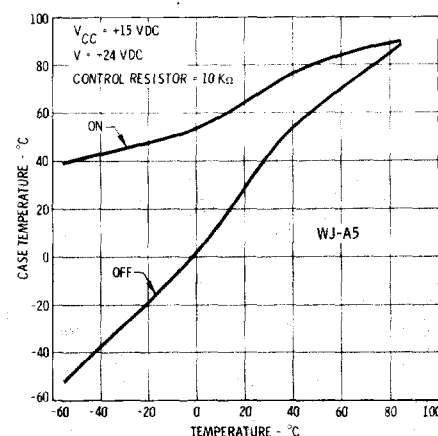


Fig. 45—Case Temperature vs. Ambient Temperature.

should be used for the A7, 30 in² for the A19 and 40 in² for the A9, to maintain less than a 10°C rise in case temperature above ambient.

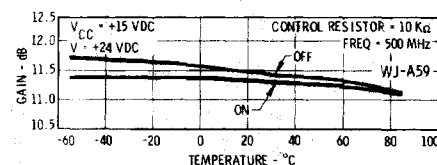


Fig. 46—Gain vs. Temperature.

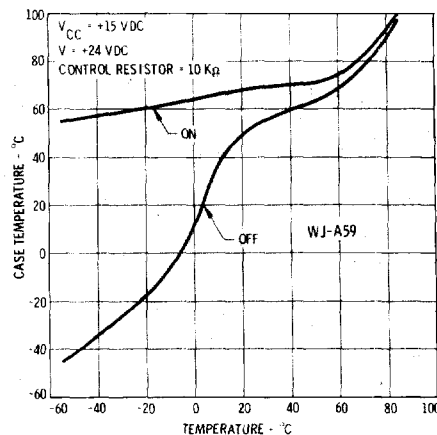


Fig. 47—Case Temperature vs. Ambient Temperature.

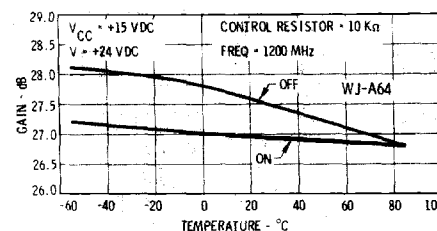


Fig. 48—Gain vs. Temperature.

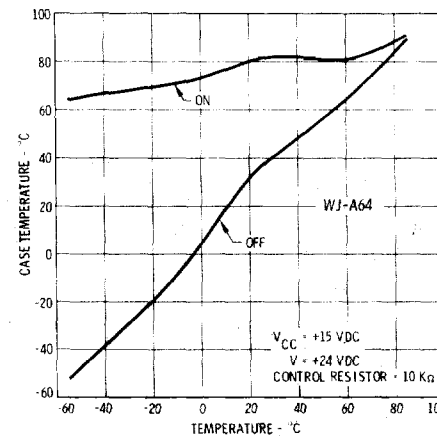


Fig. 49—Case Temperature vs. Ambient Temperature.

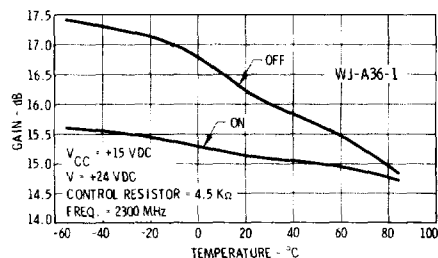


Fig. 50—Gain vs. Temperature.

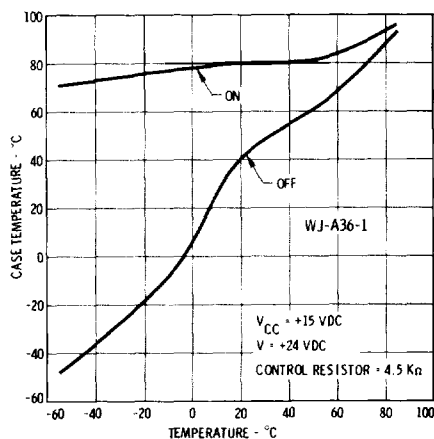


Fig. 51—Case Temperature vs. Ambient Temperature.

Case Temperature Regulation

By using the WJ-331240 proportional controller, the case temperature on the TO-8 Amplifier can be held to a much narrower range than it would be without it. This, in turn, would reduce the gain variation as a function temperature. Many of the W-J TO-8s exhibit excellent gain stability as a function of temperature, and hence would not benefit as much as the controller.

Figures 44, 46, 48, and 50 are examples of gain temperature variation as a function of ambient temperature. Figures 45, 47, 49, and 51 are examples of case temperature variation with and without the controller. For example, the WJ-A5 at its worst case frequency (500 MHz) varies approximately .5 dB over the case temperature range of -54°C to $+85^{\circ}\text{C}$ without the controller. It decreases to .3 dB with the controller as shown in Figure 44. Figure 45 shows the change in case temperature for that range. The WJ-A59 amplifier shows a similar improvement in Figure 46.

A more substantial improvement can be obtained at the high end of the band on the two-stage high frequency amps such as the A64 and A36-1. At 1200 MHz which is worst

case for the A64, the gain variation is reduced from 1.3 dB to .4 dB as shown in Figure 48. The worst case point on the A36-1 (2300 MHz) improved from 2.6 dB to .9 dB.

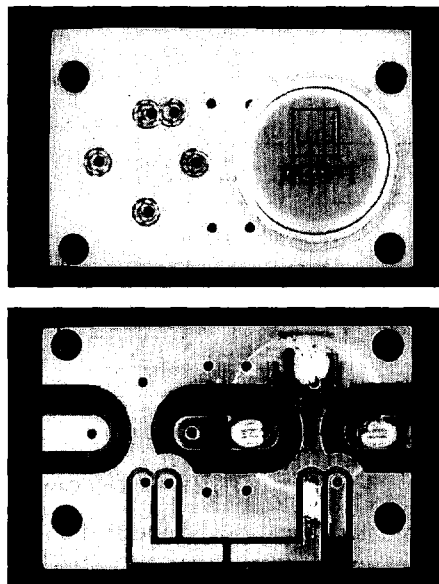


Fig. 52—Typical TO-8 Circuit Board.

Frequency Stability

Each W-J cascable amplifier has been designed to be unconditionally stable with load VSWR's of any phase angle. In order to realize inherent stability, certain installation procedures and methods must be employed.

Instability falls typically into two types: 1) Sub-band oscillations in the 100 kHz to 1 MHz region and, 2) out of-band oscillations above the frequency band. Occasionally in-band oscillations can occur, but are much more rare and are usually related to poor circuit board grounding.

Grounding

The necessity for good grounding cannot be overemphasized. The following measures provide acceptable grounding. First, the entire TO-8 case should be completely grounded across its entire surface to the circuit board. The circuit board, in turn, must be properly grounded on both sides. Shown in Figure 52 are both sides of a typical TO-8 circuit board used for mounting the TO-8's. The board ground planes are connected on the edges by the mounting screws and the case to which it is mounted. In addition, a series of small plated through-holes are placed parallel to the RF circuit trace for the

purpose of reducing the inductive reactance between the two ground planes. Insufficient ground connections can result in midband or high-end out-of-band spurious responses. The W-J TO-8 mounting kit (No. 251050-001) is the preferred technique for holding the TO-8 to the ground plane. A conductive epoxy ring that fits around the periphery of the bottom of the case is also an acceptable method, where space is at a premium.

Sub-band Instability

Much attention has been given to the problems of sub-band potential instability in recent years with improved techniques used in the design of biasing circuitry of the Watkins-Johnson line of cascable amplifiers. This effort has resulted in a vastly reduced problem in using these amplifiers with any type of load or source condition over the full operating temperature range.

It is, however, still important to add an additional 0.05 to 0.1 μf decoupling capacitor to the V_{CC} part for insured isolation from power supply related feed back. In addition, if the power supply bias lines are not properly shielded, radiation from these lines can occur causing instability. This is particularly important when high gain levels of 60 dB or higher are used.

Radiative Feedback Oscillation

Radiative feedback oscillation can occur when a high gain chain is used where the RF output voltage is in phase with the input voltage (even number of stages), and the output signal radiates back to the input because of the antenna action of both the

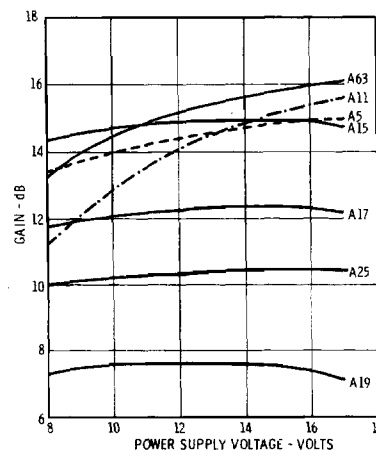


Fig. 53—Gain Versus Power Supply Voltage.

output and input circuit traces. The solution here is simply to keep output and input traces as short as possible and as far from each other as possible. For gain of over 60 dB, it is advisable to shield the input from the output by using a separate cavity for the input gain stages.

Out-of-band High Frequency Oscillations

High frequency oscillations can occur as a result of poor circuit board and amplifier case grounding. In addition, if the grounding is marginal, the units may be more susceptible to oscillation under certain load conditions. A bare lead of only one-half inch in length used to connect the RF signal from its source to the input port of the amplifier may result in oscillations. It is much better to use a properly grounded circuit board with a microstrip transmission line or coax cable. Any open lines, whether on the

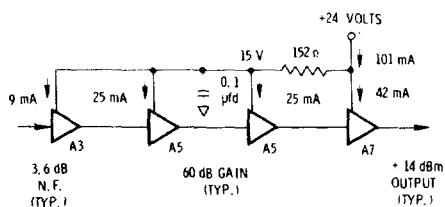


Fig. 54—Typical Four Stage Cascade.

V_{CC} port or in the RF signal path may require ferrite beads to prevent them from acting as radiators or receptors.

Power Supply Selection

The majority of Watkins-Johnson Company's cascable amplifiers operate from a nominal 15 volt DC power supply. The gain versus power supply voltage for many amplifiers is shown in Figure 53. Low level amplifiers such as the A1, A3, and A11 are represented by the curve shown for the A11 amplifier. Because of the relatively low bias level on these amplifiers, the power supply should be regulated to within 0.5 volt of the nominal 15 volt level. The higher level amplifiers such as the A5, A15, and A17 are much

¹ D. L. Cheadle, "Measure Mixer Noise Figure with Your Power Meter", *Microwaves*, March 1975.

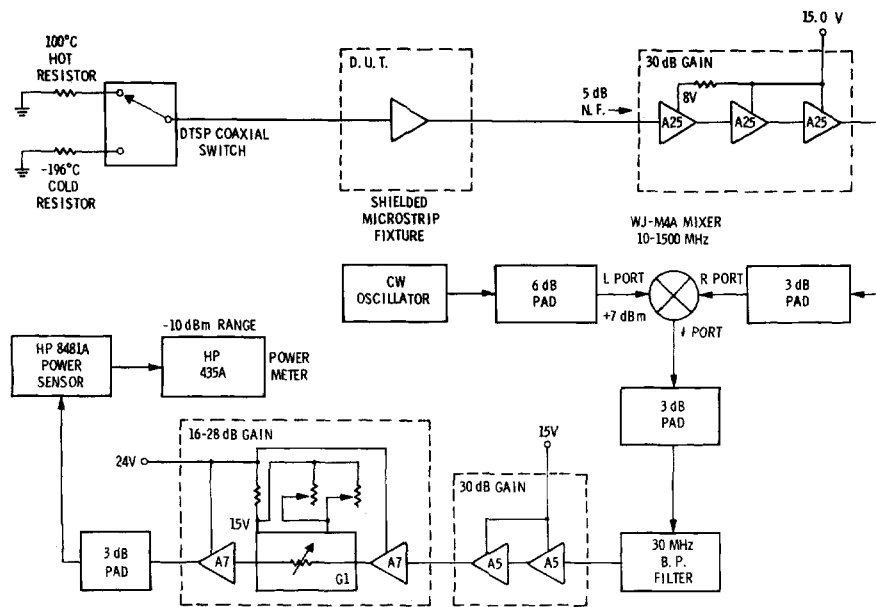


Fig. 55—Hot-Cold Cascadable Amplifier Noise Set-Up.

more tolerant and can be operated on a power supply voltage as low as 8 volts. The A15 and A25 amplifiers are extremely flat over the range of 8 to 16 volts. This extreme flatness results in an amplifier that is relatively immune to power supply ripple translating to changes in gain. In addition, 0.6 to 1.0 dB improvement in noise figure can be obtained by operating the A5, A15, or A25 at 8 volts.

Up to four stages may be cascaded in a common assembly without RF shielding between each unit. A typical four-stage assembly providing 60 dB of gain could be configured as shown in Figure 54. There is no bandwidth shrinkage when cascading the amplifiers because of the broadband feedback techniques employed.

In the assembly shown the WJ-A3 is used as the first stage followed by two WJ-A5s and one WJ-A7 as post amplifiers. The 0.1 μ F capacitor is placed across the DC bias line to suppress any tendency toward sub-band oscillation.

Noise Measurement

The noise figure that is measured on an amplifier depends to some degree on the system used to make the measurement. A few years ago the HP340B automatic system was very popular but has not been in close agreement with the more recent AILTECH type automatic system. The AIL

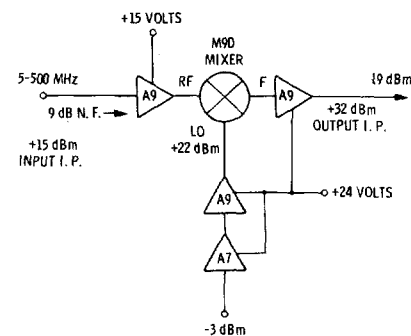


Fig. 56—High Level Mixer-Amplifier.

system, when used with a properly calibrated noise source, gives results that are typically within 0.2 dB of a hot/cold standard.

The test setup shown in Figure 55 shows one method of making NF measurements using W-J TO-8 cascable amplifiers for gain blocks. This represents a very accurate, fast production method of noise measurement using the ratio of Hot noise power to Cold noise power called the P-factor.¹

The G 1 is placed between the two A7s to provide gain adjust, which, in this case, is controlled using a coarse and fine pot for setting the reference on the power meter. The A7s are used to obtain maximum dynamic range. Using the hot resistor, a minimum of 20 dB above full scale is required by the output amplifier to